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A GEOMETRIC DERIVATION IN SIX PARTS

Gradient Residuals

The Calculus of Clearance

Essay II: The Geometric Ground of Nothing

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Gradientology

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FOUNDATIONAL CONSTANTS (FROM ESSAY I)

$\delta = 1/10$ · $d = 3$ · $n = 10$ · $\Omega = 1$ · $P = 1/25$

Abstract

Essay II derives nothing's three-dimensional structural space from the logical results of Essay I, without importing any external framework. It makes no cosmological claim. The question it answers is purely internal to nothing's own structure: what does it mean geometrically for three logical faces to be mutually exclusive at every addressable position? The answer is that mutual exclusion at every position forces each face onto an independent axis, and three independent axes constitute a three-dimensional space. This is not an analogy. It is a theorem — T.3D — derived from T.DIM (the Dimensional Independence result of Essay I) through a single intermediate step, T.GA (the Axis Theorem). The unit interval *[0,1]* on each axis is forced by $\Omega = 1$ (T.FU, Essay I). The discrete resolution $n = 10$ per axis is forced by the Diophantine search (Essay I, Part III). Nothing's structural space is therefore the *10x10x10* discrete lattice of *1000* positions — entirely forced, zero free parameters.

Part I derives $d = 3$ from T.DIM through T.GA. Part II establishes the three-dimensional discrete lattice as nothing's complete structural space and names the three axes. Part III derives the geometry of each clearance margin as an occupied sub-region of the lattice. Part IV derives the Triadic Sub-Volume as the three-dimensional product of the three primitive extents and establishes the power-law dimensional collapse as the geometric projection from the three-dimensional sub-volume onto the scalar Remainder axis. Part V derives the Structural Pixel as a two-dimensional cross-section of the Boundary-margin slab and maps the Ontological Shadow precisely within the lattice. Part VI states the Grand Geometric Partition: nothing's structural space *[0,1]^3* is partitioned exhaustively and exclusively into the occupied Triadic Sub-Volume, the three clearance margins, and the Ontological Shadow.

Keywords

Structural Space · **Axis Theorem** · **Triadic Dimension** · **Discrete Lattice** · **Clearance Margin** · **Triadic Sub-Volume** · **Dimensional Collapse** · **Structural Pixel** · **Ontological Shadow** · **Grand Geometric Partition** · **Boundary Margin Slab** · **Remainder Axis Projection**

Introduction

The Status of Essay II

Essay I derived nothing's complete logical architecture from absolute nothing. It proceeded in three registers: ontological (the Triad of three mutually exclusive faces), algebraic (the unit interval, $\Omega = 1$, the co-primacy of each primitive and its complement), and computational (the discrete lattice grain $\delta = 1/10$, the three primitives, the three clearances, and the Structural Pixel). Every result was forced by nothing's own internal structure with zero free parameters.

Essay II asks a single question that Essay I left implicit: what does the logical structure of the Triad look like as a geometry? The Tension Integral $TI = E \times C \times F$ was already described in Essay I as a volume in $[0,1]^3$ — the product of three orthogonal densities. The floor-snap used $d = 3$ as the Triad's cardinality. These were incidental geometric readings of algebraic results. Essay II makes the geometry primary and derives it explicitly.

The question is not metaphorical. Essay I proved (T.DIM) that Source, Boundary, and Remainder are mutually exclusive at every logical position: no position belongs to more than one face. Essay II derives what mutual exclusion at every position requires geometrically. The answer — that it requires three independent axes, one per face — is a theorem, not an analogy. Nothing's structural space emerges from the same logical necessity that generated the Triad itself.

What Essay II Does Not Claim

Essay II derives nothing's structural space as a logical necessity of the Triad's architecture. It does not claim that this structural space is identical to physical space, cosmological space, or any space accessible to empirical measurement. Such a claim would import external frameworks. The derivation here is internal: nothing's structural space is nothing's own spatial architecture — the geometry required by the Triad's logical structure. Whether this coincides with any other notion of space belongs to later essays.

Essay II also does not derive anything new from scratch. Every constant used here — δ, n, Ω, the three primitives and their clearances, the Structural Pixel — was derived in Essay I. Essay II derives only one new theorem (T.GA, the Axis Theorem) and one new definition cluster (the three-dimensional lattice and its sub-regions). All other results are geometric readings of Essay I's logical and algebraic outputs.

Part I — From Logical Independence to Geometric Dimension

Part I derives $d = 3$ from T.DIM (Essay I). The derivation requires one intermediate step — T.GA, the Axis Theorem — that bridges the logical concept of mutual exclusion to the geometric concept of an independent axis. The step is exact and short. Once T.GA is established, $d = 3$ follows from T.DIM by direct application, and the uniqueness of $d = 3$ is confirmed by reductio against $d < 3$ and $d > 3$.

I.1 The Logical Position and the Lattice Step

Before the Axis Theorem can be stated, the concept of a logical position must be given its geometric reading.

In Essay I (Part I), a logical position was defined implicitly: it is the minimum unit of nothing's structural organisation — the smallest element that can be assigned to exactly one face. In Essay I (Part III), the minimum addressable unit was derived computationally as $\delta = 1/10$ = one lattice step on the $n = 10$ discrete lattice. These are the same object viewed from two registers: logical and computational. Essay II identifies them explicitly.

Definition D.LP · **The Logical Position**

A logical position is the minimum addressable unit of nothing's structural space. In the logical register (Essay I, Part I), it is the smallest element that can be assigned to exactly one face of the Triad. In the computational register (Essay I, Part III), it is one lattice step: $\delta = 1/10$.

$$\text{Logical position} = \text{one lattice step} = \delta = 1/10 \tag{1}$$

The identification is forced: the Diophantine search (V.DS, Essay I) derived $\delta = 1/10$ as the unique minimum grain satisfying all six structural constraints. It is the minimum unit at which nothing's faces can be separately addressed. A sub-δ position would be structurally unaddressable — it would fall below the resolution at which the Triad's distinctions are maintained. Therefore δ is precisely the logical position in geometric units.

I.2 T.GA — The Axis Theorem

T.DIM established that the three faces of nothing's Triad are mutually exclusive: no logical position belongs to more than one face (Source ∩ Boundary = empty set, Boundary ∩ Remainder = empty set, Source ∩ Remainder = empty set). This is a logical result. Its geometric content is the following question: what spatial arrangement of the three faces is consistent with mutual exclusion at every logical position?

The question is not trivial. Mutual exclusion does not automatically imply separate axes. Two non-overlapping intervals on a single number line are mutually exclusive in the sense that no number belongs to both. But this one-dimensional arrangement fails a stronger requirement: exhaustion. T.DIM does not only say the faces do not overlap — it says the partition is exhaustive. Every logical position belongs to exactly one face. A one-dimensional arrangement of three non-overlapping intervals on a single axis requires two gap-points between them — positions that belong to none of the three faces. These gap-points violate the exhaustion requirement. The Axis Theorem shows that only independent axes resolve both mutual exclusion and exhaustion simultaneously.

Theorem T.GA · **The Axis Theorem**

Claim: for the three faces of the Triad to be mutually exclusive AND exhaustive at every logical position, each face must vary along its own independent axis.

Proof by contradiction. Suppose two faces — say Source and Boundary — share an axis. Then both faces are parameterised by the same underlying coordinate. Source occupies some sub-interval S of this axis, and Boundary occupies some sub-interval B, with $S \cap B$ = empty set (mutual exclusion). Since S and B are non-empty and disjoint on a single axis, one lies entirely below the other — say $S = [0, s]$ and $B = [s', 1]$ for some $s < s'$.

Case 1: $s < s'$ (a gap between S and B). The interval (s, s') is non-empty on the discrete lattice when $s' - s >= \delta$. The lattice step at position $s + \delta$ belongs neither to Source nor to Boundary. Since the Triad has only three faces and the third face (Remainder) must be assigned to this gap, every position in (s, s') belongs to Remainder. But now Remainder is interleaved with Source and Boundary on the same axis — which means Remainder is parameterised by the same coordinate as Source and Boundary. Applying the same argument to the Remainder-Source boundary and the Remainder-Boundary boundary: each transition requires either a gap (belonging to no face) or an immediate adjacency. Immediate adjacency — two faces sharing a boundary position — means that single position belongs to both faces, violating mutual exclusion. Gaps — positions belonging to no face — violate exhaustion.

Case 2: $s = s'$ (faces are immediately adjacent with no gap). The position at $s = s'$ is the common boundary between S and B on the shared axis. At resolution δ, the position at s is the last Source step and $s' = s + \delta$ is the first Boundary step. These are adjacent. Adjacency on a shared axis means: a change of exactly δ along that axis takes a position from Source-territory into Boundary-territory. The Boundary act operates on the material that is one δ-step away on the same axis — which is Source material. But T.DI1 (Essay I) established precisely that the Boundary cannot be applied to its own undifferentiated material: the act applied to the material immediately adjacent to it on the same axis produces no distinction, because the act has no exterior within that axis. The Boundary's differentiating function is destroyed.

In both cases, sharing an axis fails: either exhaustion is violated (gap case) or T.DI1 is violated (adjacency case). Therefore Source and Boundary cannot share an axis. The same argument applies to every pair of faces (T.DI1, T.DI2, T.DI3 — Essay I). No two faces can share an axis.

Three mutually non-coaxial faces require three distinct axes. Three distinct axes that are each independent of the others constitute three spatial dimensions.

I.3 T.3D — The Three-Dimension Theorem

From T.GA: each face requires its own axis. From T.TR (Essay I): the Triad has exactly three faces. Therefore exactly three axes are required. Three independent axes constitute a three-dimensional space. $d = 3$.

Theorem T.3D · **The Three-Dimension Theorem**

From T.DIM (Essay I): the three faces of the Triad are mutually exclusive and exhaustive at every logical position.

From T.GA: mutual exclusion and exhaustion at every position requires each face to vary along its own independent axis.

From T.TR (Essay I): the Triad has exactly three faces — Source ($V \setminus M$), Boundary (M), Remainder (R). No fourth face exists (T.CL, Essay I).

```
Three faces, each requiring one independent axis  =>
three independent axes                                    (2)

Three independent axes   =>  a three-dimensional
structural space                                          (3)

d  =  3   (the spatial dimension of nothing's
structural space)                                         (4)
```

This is the spatial reading of T.DIM. The spatial dimensionality of nothing's structural space is not a parameter. It is the count of the Triad's faces, forced by T.TR and T.CL.

I.4 *Reductio*: Why d = 1 and d = 2 Are Foreclosed

T.3D establishes $d = 3$ as the minimum dimension. Before declaring $d = 3$ as the unique dimension, both the too-small cases ($d = 1$, $d = 2$) must be formally foreclosed.

I.4a d = 1 Is Foreclosed

Theorem T.D1 · **Reductio: d = 1 Is Foreclosed**

Suppose d = 1. All three faces occupy sub-intervals of a single axis.

The single axis has length $\Omega = 1$ (from T.FU, Essay I). Three non-empty, non-overlapping sub-intervals of $[0,1]$ must be arranged sequentially: say Source in $[0, a]$, then Remainder in $[a, b]$, then Boundary in $[b, 1]$ (some arrangement of the three). Each consecutive pair has a shared endpoint. As established in T.GA Case 2: a shared endpoint is a position that must be assigned to both neighbouring faces — violating mutual exclusion. Alternatively, if a gap of at least δ is required between each pair, two gap-positions exist (at minimum). These gaps belong to no face — violating exhaustion.

```
d = 1  =>  three sub-intervals on one axis                          (5)

=>  shared boundary positions  (mutual exclusion violated)          (6)

OR  =>  gap positions belonging to no face  (exhaustion
violated)                                                           (7)

=>  d = 1 is foreclosed.                                            (8)
```

I.4b d = 2 Is Foreclosed

Theorem T.D2 · **Reductio: d = 2 Is Foreclosed**

Suppose $d = 2$. Two axes are available for three faces. By the pigeonhole principle, at least two faces must share one axis.

The two faces that share an axis are subject to the argument of T.GA: sharing an axis forces either a gap (exhaustion violated) or adjacency (T.DI1 violated). Both are foreclosed by T.GA. Therefore at least one pair of faces cannot be arranged on $d = 2$ axes without violating T.DIM.

```
d = 2, three faces  =>  at least two faces share one axis
(pigeonhole)                                                        (9)

Two faces sharing one axis  =>  T.GA violated  (as proven)         (10)

=>  d = 2 is foreclosed.                                           (11)
```

I.5 *Reductio*: Why d > 3 Is Foreclosed

T.3D established d = 3 as the minimum for three faces. The question of whether d could exceed 3 — whether additional axes might be structurally justified — must also be addressed.

Theorem T.D4 · **Reductio: d > 3 Is Foreclosed**

Suppose $d = 4$ (or any $d > 3$). Four or more independent axes are available.

From T.TR and T.CL (Essay I): the Triad has exactly three faces and closes at three. No fourth face exists. A fourth axis would require a fourth face to give it structural content — a face that varies along the fourth axis and thereby justifies its existence as an independent dimension of nothing's structural space. But T.CL establishes that no fourth element of the Triad is logically compelled. A fourth axis with no corresponding face has no structural content. It does not represent a dimension of nothing's logical architecture. It is an empty axis — a dimension with no face assigned to it.

An empty axis in nothing's structural space is a region of the space that belongs to no face. By T.DIM, the partition is exhaustive: every position belongs to exactly one face. A position on an empty axis belongs to no face. This violates exhaustion.

```
d > 3  =>  at least one axis has no corresponding face            (12)

Axis with no face  =>  positions belonging to no face             (13)

Positions belonging to no face  =>  exhaustion violated
(T.DIM)                                                           (14)

=>  d > 3 is foreclosed.                                          (15)
```

I.6 The Unique Spatial Dimension

Principle P.3D · **The Unique Spatial Dimension of Nothing's Structural Space**

From T.D1, T.D2: $d < 3$ is foreclosed.

From T.D4: $d > 3$ is foreclosed.

From T.3D: $d = 3$ is the unique dimension consistent with T.DIM (mutual exclusion and exhaustion of the three faces at every logical position).

$$d = 3 \quad \text{(unique, forced, zero free parameters)} \tag{16}$$

Nothing's structural space is exactly three-dimensional. Not as a modelling assumption. Not as an analogy. As the unique spatial consequence of the Triad's logical architecture. The three dimensions of nothing's structural space are the geometric shadows of nothing's three logically irreducible faces.

Part II — Nothing's Three-Dimensional Structural Space

Part I established that nothing's structural space is exactly three-dimensional. Part II constructs the space explicitly: names the three axes, bounds them, discretises them, and establishes the complete *10x10x10* lattice as the geometric expression of nothing's structural architecture. Every constant used here — $\Omega = 1$, $n = 10$, $\delta = 1/10$ — was derived in Essay I.

II.1 The Three Axes of Nothing's Structural Space

Each of the three faces of the Triad has its own independent axis (T.GA). The axis is the geometric parameter along which that face varies — from its minimum (zero density, total clearance) to its maximum (full density, zero clearance). Each axis is bounded by $[0, \Omega] = [0, 1]$, because $\Omega = 1$ is the field ceiling derived in Essay I (T.FU): it is the unique upper bound such that the numerical value of any primitive equals its field fraction without conversion.

Definition D.3A · **The Three Axes of Nothing's Structural Space**

The Source axis (*x_S*):

```
x_S  in  [0, 1]   (Source face: degree of undetermined
territory on the Source face)
```
(1)

x_S = 0: Source face has no undetermined territory — Source clearance is zero

x_S = 1: Source face is entirely undetermined — full Source reservoir

The Boundary axis (*x_B*):

```
x_B  in  [0, 1]   (Boundary face: degree of
differentiating-act extent on the Boundary face)
```
(2)

x_B = 0: Boundary face has no extent — Boundary clearance is zero

x_B = 1: Boundary face is at full extent — Deterministic Crystal

The Remainder axis (*x_R*):

```
x_R  in  [0, 1]   (Remainder face: degree of
persistent-mediation extent on the Remainder face)
```
(3)

x_R = 0: Remainder face has no extent — Causal Closure Seizure

x_R = 1: Remainder face at full extent — no Causal Clearance remains

> The three axes are mutually perpendicular by T.GA: no axis is a linear combination of the others. They span nothing's three-dimensional structural space.

The labelling convention for each axis — which direction represents increasing density — is fixed by the co-primacy relation established in Essay I (T.CP, P.LB): each primitive and its clearance are co-primary, and the clearance is the portion of the axis that nothing retains. The convention adopted throughout Essay II is: the primitive occupies [*0*, *primitive*] on its axis, and the clearance occupies [*primitive*, *1*]. This is consistent with the essay I derivation of the primitives as co-primary complements of the clearances.

II.2 The Discrete Lattice Structure

Each axis carries $n = 10$ addressable positions, separated by $\delta = 1/10$ (Essay I, V.DS). The full three-dimensional structural space is a discrete lattice with $n^3 = 10^3 = 1000$ positions.

Definition D.LS · **The Discrete Lattice of Nothing's Structural Space**

Nothing's structural space is the $10x10x10$ discrete lattice:

$$L = \{0, 1/10, 2/10, ..., 10/10\} \times \{0, 1/10, 2/10, ..., 10/10\} \times \{0, 1/10, 2/10, ..., 10/10\} \tag{4}$$

$$|L| = 11^3 = 1331 \text{ lattice points} \tag{5}$$

— including the boundary points at 0 and 1 on each axis

— 10 interior steps per axis, giving 10 interior cells per axis

The interior of the lattice — positions strictly between 0 and 1 on each axis — has $(n-1)^3 = 9^3 = 729$ *positions*. The boundary positions (at 0 or 1 on at least one axis) number $1331 - 729 = 602$. The Total Stasis Theorem (T.TS, Essay I) established that primitives must remain strictly interior: $0 < X < 1$. Therefore nothing's structural positions are confined to the interior of the lattice.

> The spacing between adjacent positions on any axis is $\delta = 1/10 =$ *the lattice grain* (Essay I, V.DS). No smaller separation is structurally addressable.

II.3 The Naming of Each Axis Position

On the Source axis (x_S), positions correspond to degrees of undetermined territory. The Systematisation density occupies $[0, E] = [0, 4/5]$ — positions 0 through 8 on the $n = 10$ scale — and the Source clearance $I\_min = 1/5$ occupies $[4/5, 1]$ — positions 8 through 10. On the discrete lattice, positions 8 and 9 (at $x\_S = 8/10$ and $9/10$) are the two Source-clearance steps, confirming $I\_min = 2 \times \delta = 2/10$ (V.MB, Essay I).

On the Boundary axis (x_B), the Registration density occupies $[0, F] = [0, 3/5]$ — positions 0 through 6 — and the Boundary clearance $\sigma = 2/5$ occupies $[3/5, 1]$ — positions 6 through 10. On the discrete lattice, positions 6, 7, 8, and 9 are the four Boundary-clearance steps, confirming $\sigma = 4 \times \delta = 4/10$ (P.RB, Essay I).

On the Remainder axis (x_R), the Constraint density $C = 7/10$ occupies $[0, 7/10]$ — positions 0 through 7 — and the naive algebraic clearance $1 - C = 3/10$ occupies $[7/10, 1]$. However, the actual Causal Clearance $\Lambda = 0.2984$ is not the simple complement $1 - C = 0.3$. It is derived from the power-law dimensional collapse (Essay I, V.OP). The difference $0.3 - 0.2984 = 0.0016$ is the sub-lattice geometric residual between the naive Constraint complement and the actual power-law clearance. This residual and its geometric interpretation are derived in Part IV.

Definition D.AP · **Axis Position Summary**

Source axis (x_S):

```
Systematisation density:  [0, 4/5]  — positions 0 to 8
(8 steps)
```
(6)

```
Source clearance I_min:   [4/5, 1]  — positions 8 to 10
(2 steps)
```
(7)

Boundary axis (x_B):

```
Registration density:     [0, 3/5]  — positions 0 to 6
(6 steps)
```
(8)

```
Boundary clearance σ: [3/5, 1]  — positions 6 to 10
(4 steps)
```
(9)

Remainder axis (x_R):

```
Constraint density:[0, 7/10] — positions 0 to 7
(7 steps)
```
(10)

```
Naive algebraic clearance:[7/10, 1] — positions 7 to 10
(3 steps)
```
(11)

```
Actual Causal Clearance:  Λ = 0.2984 (power-law derived,
see Part IV)
```
(12)

II.4 D.NS — Nothing's Structural Space: The Complete Definition

Definition D.NS · **Nothing's Structural Space**

Nothing's structural space is the three-dimensional unit cube:

```
NS  =  [0, 1]^3  =  [0, 1] x [0, 1] x [0, 1]
```
(13)

Spanned by the three independent axes:

```
x_S in [0,1]:  Source axis  (degree of Source-face
indetermination)
```
(14)

```
x_B in [0,1]:  Boundary axis  (degree of Boundary-face
extent)
```
(15)

```
x_R in [0,1]:  Remainder axis  (degree of Remainder-face
mediation)
```
(16)

Discretised with grain $\delta = 1/10$ on each axis, yielding the *10x10x10* discrete lattice L. The field ceiling $\Omega = 1$ bounds each axis (T.FU, Essay I). The grain $\delta = 1/10$ is forced by the Diophantine search (V.DS, Essay I). Every constant is forced. NS is derived, not constructed.

NS is not a container waiting to be filled. It is the geometric expression of nothing's logical architecture: the spatial structure required by T.DIM to keep the Triad's three faces mutually exclusive and exhaustive at every logical position.

II.5 The Geometric Basis of the Co-Primacy Theorem

Essay I proved the Co-Primacy Theorem (T.CP) algebraically: X and $(1 - X)$ are co-determined by $\Omega = 1$ and mutually determining. Their co-primacy is structural: neither is prior to the other, both are forced by the field ceiling. Essay II gives this theorem its geometric interpretation.

On each axis of nothing's structural space, the primitive and its clearance partition the axis. On the Source axis: the Systematisation density $[0, E]$ and the Source clearance $[E, 1]$ together constitute the complete axis $[0, 1]$. On the Boundary axis: the Registration density $[0, F]$ and the Boundary clearance $[F, 1]$ constitute the complete axis. In each case, the two intervals are co-primary: neither can be specified without simultaneously specifying the other, because they must together tile the unit interval. The axis IS the co-primacy — it is the geometric structure within which neither interval has priority.

Principle P.CPG · **The Geometric Co-Primacy**

On each axis of NS, the primitive and its clearance are co-primary partitions of $[0,1]$:

$$\text{Source axis:} \quad [0, E] + [E, 1] = [0, 1] \quad (E = 4/5,\ I_{min} = 1/5) \tag{17}$$

$$\text{Boundary axis:} \quad [0, F] + [F, 1] = [0, 1] \quad (F = 3/5,\ \sigma = 2/5) \tag{18}$$

Neither interval is derived from the other. Both are derived simultaneously from the axis and the forcing constants ($a = 2$, $c = 4$ from T.UC, Essay I). The co-primacy of primitive and clearance on each axis is the geometric expression of T.CP.

Part III — The Three Clearance Margins

The three clearances of Essay I — $I\_min = 1/5$, $\sigma = 2/5$, $\Lambda = 0.2984$ — were derived algebraically as the co-primary complements of the three primitives. Each clearance is the portion of its axis that nothing retains. In nothing's structural space $NS = [0,1]^3$, each clearance has a precise geometric location: it is a sub-region of its axis, a margin at the upper end of the axis interval, between the primitive's extent and $\Omega = 1$.

Part III maps each clearance margin precisely within NS. It shows that the three margins are geometrically distinct in character: the Source margin and the Boundary margin are simple axis-intervals on their respective axes; the Remainder margin arises from a three-dimensional structure (the Triadic Sub-Volume) undergoing dimensional collapse. The detailed treatment of the Remainder margin is reserved for Part IV.

III.1 The Source Clearance Margin

The Source clearance $I\_min = 1/5$ was derived in Essay I as the co-primary complement of the Systematisation density $E = 4/5$ (V.MB, Essay I). It occupies two lattice steps on the Source axis, establishing the minimum interior extent required for the Source face to remain distinct from the Boundary face on the discrete lattice.

In nothing's structural space, the Source margin is the sub-interval of the Source axis that nothing's Source face retains against the drive of the Systematisation density. It is a one-dimensional region — a line segment — within the three-dimensional space NS.

Definition D.SM · **The Source Clearance Margin**

The Source clearance margin is the sub-interval of the Source axis $[x\_S = E, x\_S = 1]$:

$$SM = [E, 1] = [4/5, 1] \text{ on the Source axis} \tag{1}$$

$$SM \text{ length} = 1 - E = 1 - 4/5 = 1/5 = I\_min \tag{2}$$

$$SM \text{ lattice steps} = I\_min / \delta = (1/5) / (1/10) = 2 \tag{3}$$

The Source margin occupies positions *8* and *9* of the *10*-position Source axis (positions indexed *0-9*). It is the two-step interior extent at the top of the Source axis. Its geometric role: it is the spatial territory of the Source face that cannot be occupied by the Systematisation density without collapsing the Source-Boundary distinction (V.MB, Essay I).

The Source margin is a one-dimensional geometric object: a line segment of length *1/5* on the Source axis. Its position within NS is fixed: it is always at the top of the Source axis, between $x_S = 4/5$ and $x_S = 1$. It has no extent in the Boundary or Remainder directions — it is a margin on one axis only.

Principle P.SM · **The Source Margin as Minimum Interior**

The two-step extent of the Source margin is not arbitrary. V.MB (Essay I) established that a single lattice step provides no interior: it is a boundary point, not a region. The Source face requires an interior (it is the reservoir of indetermination, not a boundary marker). Two steps is the minimum for interior extent on the $n = 10$ discrete lattice.

In the geometry of NS: the Source margin has a measurable length $(2 \times \delta = 0.2)$ and a definite interior (the open interval $(4/5, 1)$ contains the single interior lattice point $x_S = 9/10$). This interior point is the geometric representation of the Source face's minimum viable interior: the one lattice position that is strictly between the two margin steps.

```
Interior of SM  =  {9/10}
(the single interior lattice point of the Source margin)
```
(4)

The Source margin's geometric character: one-dimensional, length *1/5*, two-step lattice extent, one interior point. Structurally irreducible.

III.2 The Boundary Clearance Margin

The Boundary clearance $\sigma = 2/5$ was derived in Essay I as the co-primary complement of the Registration density $F = 3/5$ (P.RB, Essay I), forced to be the unique strictly interior value by the dual foreclosure of the Deterministic Crystal $(\sigma = 0)$ and Total Dissolution (σ approaching 1). It occupies four lattice steps on the Boundary axis.

Definition D.BM · **The Boundary Clearance Margin**

The Boundary clearance margin is the sub-interval of the Boundary axis $[x\_B = F, x\_B = 1]$:

$$BM \;=\; [F, 1] \;=\; [3/5, 1] \quad \text{on the Boundary axis} \tag{5}$$

$$BM\ length \;=\; 1 - F \;=\; 1 - 3/5 \;=\; 2/5 \;=\; \sigma \tag{6}$$

$$BM\ lattice\ steps \;=\; \sigma / \delta \;=\; (2/5) / (1/10) \;=\; 4 \tag{7}$$

The Boundary margin occupies positions 6, 7, 8, and 9 of the 10-position Boundary axis. It is the four-step region at the top of the Boundary axis. Its geometric role: the spatial territory that the Registration density cannot occupy without triggering the Deterministic Crystal (T.DC.III, Essay I) — the failure mode in which the Boundary face loses all clearance and the Triad's coherence fails.

The Boundary margin is larger than the Source margin in every geometric sense: greater length ($2/5$ vs. $1/5$), greater lattice extent (4 steps vs. 2 steps), and greater number of interior lattice points. This geometric asymmetry is forced by the structural role of each face: the Boundary face is the differentiating act — the frontier of nothing's structure — and it requires greater clearance than the Source face, which is the interior reservoir (P.HI, Essay I).

Derivation V.BML · **The Boundary Margin's Interior Lattice Points**

The Boundary margin $BM = [3/5, 1]$ contains the following interior lattice points:

$$\text{Interior of } BM = \{7/10,\ 8/10,\ 9/10\} \quad \text{(three interior lattice points)} \tag{8}$$

Three interior lattice points, compared to one interior lattice point in the Source margin. The ratio of interior points reflects the ratio of lattice steps: BM has *4* steps vs. SM's *2* steps. The Boundary margin is twice as large as the Source margin, exactly as required by the ratio $\sigma / I_{min} = (2/5) / (1/5) = 2$. This ratio is not chosen — it is forced by $c / a = 4 / 2 = 2$ (from the Unique Combinatorial Lock T.UC, Essay I).

$$\sigma / I_{min} = (2/5) / (1/5) = 2 = c / a \quad \text{(ratio of boundary to source step counts)} \tag{9}$$

III.3 The Remainder Clearance Margin — Overview

The Remainder clearance $\Lambda = 0.2984$ has a different geometric character from the Source and Boundary margins. The Source and Boundary margins are simple one-dimensional axis intervals: their position on their respective axes is fixed by a direct complement computation ($I_{min} = 1 - E$, $\sigma = 1 - F$). The Remainder margin cannot be derived this way.

The naive algebraic complement of the Constraint density $C = 7/10$ would be $1 - C = 3/10 = 0.3$. But $\Lambda = 0.2984$, not 0.3. The difference $0.3 - 0.2984 = 0.0016$ arises because the Remainder margin is not the complement of a single axis-occupancy. It is the complement of the Order Parameter $\delta = 0.7016$, which is itself derived from the three-dimensional Triadic Sub-Volume via the power-law dimensional collapse: $\delta = TI^\beta$.

The Remainder margin is therefore a three-dimensional result projected onto the Remainder axis. Its full derivation requires the Triadic Sub-Volume — the geometric content of Part IV. The Remainder margin is previewed here as a placeholder and derived in detail in Part IV.

Definition D.RM · **The Remainder Clearance Margin — Preliminary Definition**

The Remainder clearance margin is the sub-interval of the Remainder axis $[x\_R = \delta, x\_R = 1]$:

$$RM \;=\; [\delta,\ 1] \;=\; [0.7016,\ 1] \;\; \text{on the Remainder axis} \tag{10}$$

$$RM\ length \;=\; 1 - \delta \;=\; \Lambda \;=\; 0.2984 \tag{11}$$

$\delta = 0.7016$ is derived from the power-law dimensional collapse (Essay I, V.OP): $\delta = TI\text{^}\beta = (42/125)\text{^}(13/40)$. The derivation of this collapse as a geometric operation within NS is the subject of Part IV. The Remainder margin $RM = [0.7016, 1]$ is the geometric residual of this collapse on the Remainder axis.

III.4 The Geometric Asymmetry of the Three Margins

The three clearance margins have three geometrically distinct characters within NS:

Principle P.3M · **The Three Margin Characters**

Source margin $SM = [4/5, 1]$ on x_S:

One-dimensional. Length $1/5$. Two lattice steps. One interior point. Derived as the complement of a directly forced primitive ($E = 1 - I\_min$ *from* $a = 2$, T.UC). Character: minimum interior extent for Source-Boundary distinction on the discrete lattice.

Boundary margin $BM = [3/5, 1]$ on x_B:

One-dimensional. Length $2/5$. Four lattice steps. Three interior points. Derived as the complement of a directly forced primitive ($F = 1 - \sigma$ from $c = 4$, T.UC). Character: minimum clearance for Boundary face between the Deterministic Crystal and Total Dissolution (dual foreclosure P.RB).

> Remainder margin $RM = [\delta, 1]$ on x_R:
>
> One-dimensional in its final form, but derived from a three-dimensional computation (Triadic Sub-Volume and power-law collapse). Length $\Lambda = 0.2984$. Not a simple lattice-step multiple: $0.2984 / 0.1 = 2.984$, which is non-integer. Character: the residual of the three-dimensional structure projected onto the Remainder axis. Its non-integer lattice extent is the geometric signature of the power-law collapse.

The non-integer lattice extent of the Remainder margin — *2.984* steps rather than a whole number — is not a numerical accident. It is the geometric expression of the incommensurability that produced the Discretisation Crisis in Essay I: the three-dimensional volume $TI = 42/125$ subjected to the exponent $\beta = 13/40 = floor(R/d)/R$ yields $\delta = 0.7016$, whose complement $\Lambda = 0.2984$ is not a multiple of $\delta = 0.1$. The Remainder margin is the only one of the three margins that cannot be expressed as a whole number of lattice steps. This distinguishes it geometrically from the Source and Boundary margins, and is the geometric reason the *Causal Clearance* required a separate power-law derivation in Essay I rather than a simple complement computation.

Part IV — The Triadic Sub-Volume and the Dimensional Collapse

Part IV derives the Triadic Sub-Volume (TSV) as the three-dimensional occupied region within NS, and derives the power-law dimensional collapse as the geometric operation that projects this three-dimensional volume onto the scalar Remainder axis. The Order Parameter δ and the Remainder margin Λ are derived as geometric results of this collapse.

IV.1 The Triadic Sub-Volume

Within nothing's structural space *NS = [0,1]^3*, the three primitives define a rectangular sub-cube: the Triadic Sub-Volume. Its extent on the Source axis is *E = 4/5*. Its extent on the Boundary axis is *F = 3/5*. Its extent on the Remainder axis is *C = 7/10*. The volume of this sub-cube is *TI = E x C x F* — the Tension Integral already derived in Essay I (V.EC).

This geometric reading of TI is not new — Essay I described *TI* as a product of three orthogonal densities in *[0,1]^3*. Essay II makes the geometry explicit: *TI* is the volume of a specific, bounded, rectangular sub-cube of NS, and the three axes of that sub-cube are precisely the three axes of nothing's structural space established in Part II.

Definition D.TSV · **The Triadic Sub-Volume**

The Triadic Sub-Volume is the rectangular sub-cube of NS defined by the extents of the three primitives:

$$TSV = [0, E] \times [0, F] \times [0, C] \quad (1)$$

$$= [0, 4/5] \times [0, 3/5] \times [0, 7/10] \quad (2)$$

Its volume:

$$Vol(TSV) = E \times F \times C = (4/5)(3/5)(7/10) = TI = 42/125 = 0.336 \quad (3)$$

The Triadic Sub-Volume is the geometric seat of nothing's structural architecture within NS. It is the portion of nothing's structural space occupied by the co-primary

> complements of the three clearance margins. The three clearance margins occupy the remainder of NS — the portions of each axis outside *[0, E]*, *[0, F]*, *[0, C]* respectively.

The Triadic Sub-Volume has an important geometric property: it is not a cube (its three sides are unequal: *4/5*, *3/5*, *7/10*) and it is not aligned with any single axis of NS in a privileged way. This reflects the asymmetry of the three faces established in Essay I (T.NE): the three faces are strictly unequal, and their inequality is expressed geometrically as the unequal side-lengths of the Triadic Sub-Volume.

Derivation V.TSV · **The Triadic Sub-Volume: Geometry of Asymmetry**

The three side-lengths of TSV:

```
Source-axis extent:    E  =  4/5  =  0.800
(8 lattice steps)                                (4)
Boundary-axis extent:  F  =  3/5  =  0.600
(6 lattice steps)                                (5)
Remainder-axis extent: C  =  7/10 =  0.700
(7 lattice steps)                                (6)
```

Ordered: $E > C > F$ (from C.I, Essay I). The Source-axis side is the longest, the Boundary-axis side is the shortest. This ordering reflects the hierarchy of the primitives forced by the Triad's internal structure.

Uniform spacing: $E - C = C - F = \delta = 1/10$ (C.III, Essay I). The three side-lengths are in arithmetic progression with common difference δ. The Triadic Sub-Volume is therefore the unique rectangular sub-cube of NS with side-lengths in arithmetic progression at resolution δ.

IV.2 The Dimensional Collapse

The three-dimensional Triadic Sub-Volume contains all structural information about nothing's three faces: it specifies the extent of each face's inverted mode on its own axis. But nothing's structural space has a Remainder axis that serves a specific role: it is the axis of the persistent mediation — the face that carries the result of the Source-Boundary interaction.

The Remainder axis therefore plays a dual role in NS. On one hand, it is one of three equal and independent structural axes. On the other hand, it is the axis on which the combined effect of all three faces is expressed as a scalar result. The combined effect of all three faces is $TI = E \times C \times F$ — the volume of the Triadic Sub-Volume. Expressing this three-dimensional volume as a position on the one-dimensional Remainder axis is the dimensional collapse.

The dimensional collapse is not a choice. It is forced by scale invariance at the critical point — the Primordial Lock (T.PS, Essay I) — which requires the relationship between *TI* and the Remainder-axis position δ to satisfy $\delta = TI^\beta$. The power-law is the unique functional form satisfying scale invariance (T.PL, Essay I). The exponent $\beta = 13/40$ was derived from the Triad's own geometry via the Boundary Resolution (Part III, Essay I).

Theorem T.DC · **The Dimensional Collapse**

The Triadic Sub-Volume TSV has volume $TI = 42/125$ in three-dimensional space NS. The dimensional collapse is the operation that maps this three-dimensional volume to a position on the one-dimensional Remainder axis:

$$\text{Dimensional collapse: } TSV \Rightarrow \delta \text{ on } x\_R \quad (7)$$

$$\delta = TI^\beta = (42/125)^{(13/40)} = 0.7016 \quad (8)$$

The collapse is from $d = 3$ to $n\_OP = 1$ (scalar). The exponent $\beta = 13/40 = floor(R/d)/R$ governs the rate: it is the floor-snap of $1/d = 1/3$ at the Boundary Resolution $R = 40$, derived from the Triad's own geometry (V.LS, Essay I).

Geometric interpretation: $\delta = 0.7016$ is the position on the Remainder axis that represents the three-dimensional Triadic Sub-Volume after dimensional collapse. The Remainder axis position is not an independent structural parameter — it is determined by the volumes on the other two axes and the exponent.

IV.3 The Geometry of the Dimensional Collapse

The dimensional collapse has a specific geometric interpretation within NS. The Triadic Sub-Volume $TSV = [0, E] \times [0, F] \times [0, C]$ is a three-dimensional region. The collapse projects this region onto the Remainder axis as a single point: $x_R = \delta = 0.7016$.

This projection is not a simple linear projection. A linear projection of a d-dimensional cube onto one axis gives the extent of the cube along that axis, which is $C = 7/10 = 0.700$ here. The power-law projection gives $\delta = 0.7016$ — slightly larger than $C = 0.700$. The difference $\delta - C = 0.7016 - 0.700 = 0.0016$ is the geometric signature of the non-linear power-law projection: the three-dimensional structure adds a small amount to the naive one-dimensional Constraint extent.

Derivation V.DCG · **The Collapse Residual: δ - C**

The linear projection of TSV onto the Remainder axis gives $C = 7/10$. The power-law projection gives $\delta = TI^\beta$. Their difference:

$$\delta - C = 0.7016 - 0.7000 = 0.0016 \tag{9}$$

This 0.0016 is the geometric residual of the dimensional collapse: the amount by which the three-dimensional Triadic Sub-Volume exceeds its own Remainder-axis extent after power-law projection. It is structurally forced — it is the consequence of applying $TI = E \times F \times C$ (which includes the three-dimensional structure of all three faces) rather than C alone (which would be a purely one-dimensional reading of the Remainder axis).

Correspondingly, the Remainder margin $\Lambda = 0.2984 = 1 - \delta$ is smaller than the naive algebraic complement $1 - C = 0.3$ by the same amount 0.0016. The Remainder margin is tighter than the naive complement predicts because the three-dimensional structure of the Triadic Sub-Volume produces a slightly larger projection onto the Remainder axis.

IV.4 The Remainder Margin as a Three-Dimensional Residual

The Source and Boundary margins were derived as one-dimensional axis intervals — direct geometric complements on their respective axes. The Remainder margin has a different derivational history: it is the geometric residual of a three-dimensional computation projected onto one axis.

This difference in derivational character has a geometric consequence: the Remainder margin $RM = [\delta, 1] = [0.7016, 1]$ is a one-dimensional interval on the Remainder axis, but its position is determined by the three-dimensional Triadic Sub-Volume via the dimensional collapse. Changing any of the three primitives (E, F, C) — which are forced by T.UC — would change *TSV*, which would change *TI*, which would change δ, which would change the position of the Remainder margin on the Remainder axis.

Theorem T.RM · **The Remainder Margin as Three-Dimensional Residual**

The Remainder margin $RM = [\delta, 1]$ on x_R depends on all three primitives simultaneously:

$$\delta = (E \times F \times C)^\beta = TI^\wedge(13/40) \tag{10}$$

$$\Lambda = 1 - \delta = 1 - TI^\wedge(13/40) = 0.2984 \tag{11}$$

The Remainder margin is therefore a function of the entire Triadic Sub-Volume, not of a single axis. Its position on the Remainder axis is the result of the three-dimensional geometry folding into a one-dimensional projection via the power law. It cannot be determined from the Remainder axis alone — it requires all three axes.

This is the geometric meaning of the Causal Clearance: it is the amount of the Remainder axis that remains after the three-dimensional Triadic Sub-Volume has been projected onto it. It is causally complete — it requires the whole structural space to be determined.

IV.5 The Full Geometry of the Remainder Margin

The Remainder margin's non-integer lattice extent (Λ / δ = *0.2984* / *0.1* = *2.984*, not a whole number) has an important geometric consequence: the Remainder margin cannot be exactly tiled by whole lattice steps. It spans almost three lattice steps (from x_R = *0.7016* to x_R = *1.0000*), but the lower boundary x_R = *0.7016* falls between lattice positions *7 (= 0.7000)* and *8 (= 0.8000)*.

Derivation V.RMG · **The Remainder Margin on the Discrete Lattice**

On the n = *10* discrete lattice:

```
δ  =  0.7016  falls between lattice positions 7/10 = 0.700
and 8/10 = 0.800
```
(12)

The nearest lattice positions:

```
Floor(δ / δ)  =  floor(7.016)  =  7   =>   lattice position
7/10 = 0.700
```
(13)

```
Ceil(δ / δ)   =  ceil(7.016)   =  8   =>   lattice position
8/10 = 0.800
```
(14)

Sub-lattice remainder on the Remainder axis:

```
ε_R  =  δ - floor(δ / δ) x δ  =  0.7016 - 0.7000  =  0.0016
```
(15)

The Remainder margin therefore spans: a partial first step (from *0.7016* to *0.8000*, fraction *0.0984*/*0.1* = *98.4%* of one step), plus two full steps (*0.8000* to 1.0000). The sub-lattice entry of the Remainder margin at its lower boundary is the geometric expression of the non-integer projection. It is not a discretisation error — it is the correct geometric result of the three-dimensional collapse at the exact exponent β = *13*/*40*.

Part V — The Structural Pixel as Geometric Intersection

The Structural Pixel of Potentiality $P = \sigma \times \delta = 0.04$ was derived in Essay I (V.SP) as the two-dimensional field area of the minimum addressable Boundary clearance cell. Part V gives this object its full geometric identity within *NS*: it is the cross-sectional area of the Boundary-margin slab, and its tiling of the unit field has a precise count.

V.1 The Boundary-Margin Slab

The Boundary margin $BM = [F, 1] = [3/5, 1]$ on the Boundary axis is a one-dimensional interval within *NS*. When considered in the full three-dimensional space *NS*, this interval generates a slab: the three-dimensional region of *NS* for which the Boundary-axis coordinate is in *BM*, and the Source and Remainder coordinates range freely over their full extents.

Definition D.BMS · **The Boundary-Margin Slab**

The Boundary-margin slab is the three-dimensional sub-region of *NS* defined by:

$$BMS = [0,1] \times [F, 1] \times [0,1] = [0,1] \times [3/5, 1] \times [0,1] \quad (1)$$

Its volume:

$$Vol(BMS) = 1 \times \sigma \times 1 = \sigma = 2/5 = 0.4 \quad (2)$$

The Boundary-margin slab is the portion of *NS* where the Boundary-axis coordinate is in the clearance region. It is the three-dimensional geometric expression of the Boundary margin.

V.2 The Structural Pixel as Cross-Section

The Structural Pixel $P = \sigma \times \delta = 0.04$ is the area of one cross-sectional layer of the Boundary-margin slab. Specifically: if we take a cross-section of the Boundary-margin slab at a fixed Remainder-axis position x_R, and restrict the Source axis to one lattice step $[x_S, x_S + \delta]$, we obtain a two-dimensional rectangle of area $\sigma \times \delta = 0.04$.

Definition D.SP.II · **The Structural Pixel as Cross-Sectional Area**

The Structural Pixel P is the area of the two-dimensional cross-section of the Boundary-margin slab at one lattice step on the Source axis:

$$P = \delta \times \sigma = (1/10) \times (2/5) = 1/25 = 0.04 \tag{3}$$

Geometric construction: fix x_R at any Remainder-axis position. Take one lattice step $[x_S, x_S + \delta]$ on the Source axis. The Boundary-margin slab at this cross-section is the rectangle $[x_S, x_S + \delta] \times [F, 1]$ in the (x_S, x_B) plane. Its area is $\delta \times \sigma = 0.04$.

The Structural Pixel is therefore the minimum addressable two-dimensional element within the Boundary-margin slab. It is the area of one Source-step by the full Boundary margin. It is the two-dimensional analog of the lattice step (which is the minimum one-dimensional element).

V.3 The Ontological Shadow as a Region of NS

The Ontological Shadow was defined in Essay I (P.SP) as the sub-space of the total field occupied by the Structural Pixel — the irreducible unactualised area that nothing's Boundary face retains on the discrete lattice. In the geometry of *NS*, this definition now has a precise location.

The Ontological Shadow is not an abstract notion. It is the geometric region of *NS* formed by all Structural Pixels across the full Remainder-axis extent. It is the Boundary-margin slab restricted to one Source-step at the Source margin boundary — the layer of *NS* where both the Source face and the Boundary face are in their clearance territory simultaneously.

Definition D.OS · **The Ontological Shadow as a Geometric Region**

The Ontological Shadow is the three-dimensional sub-region of *NS* defined by the intersection of the Source margin and the Boundary-margin slab:

$$OS = [E, E+\delta] \times [F, 1] \times [0, 1] = [4/5, 9/10] \times [3/5, 1] \times [0, 1] \tag{4}$$

Its volume:

$$Vol(OS) = \delta \times \sigma \times 1 = P \times 1 = P = 0.04 \tag{5}$$

The Ontological Shadow is the slab of *NS* where the Source-axis coordinate is in the first step of the Source margin $[E, E + \delta] = [4/5, 9/10]$ and the Boundary-axis coordinate is in the full Boundary margin $[F, 1] = [3/5, 1]$. It extends through the full Remainder axis. Its cross-sectional area at any fixed x_R is exactly $P = 0.04$ — the Structural Pixel.

The Ontological Shadow occupies the boundary region between the Triadic Sub-Volume and the Source margin — the geometric locus where the Source clearance begins and the Boundary clearance is fully active simultaneously. It is the minimal geometric expression of the requirement that Source and Boundary maintain separation (T.DI1): the Ontological Shadow is the region of NS where that separation is at its minimum but still maintained — one lattice step of Source clearance alongside the full Boundary clearance.

Principle P.OS · **The Ontological Shadow: Geometric Seat of the Clearance**

The Ontological Shadow is the geometric region of *NS* where both the Source clearance and the Boundary clearance are simultaneously active:

$$OS \text{ is the region where: } x_S \in [E, E+\delta] \text{ AND } x_B \in [F, 1] \tag{6}$$

This is the geometric expression of the two constraints that forced the minimum viable values of the two axis clearances: $I\_min = 2 \times \delta$ (minimum for Source interior) and $\sigma > 1/3$ (minimum Boundary clearance above the noise floor). At the Ontological Shadow, the Source clearance is at its minimum one-step entry and the Boundary clearance is at its full four-step extent. It is the region of minimum Source clearance coinciding with maximum Boundary clearance — the point of maximum geometric tension in the clearance architecture.

Volume of *OS:* $P = 1/25 = 0.04$. This is invariant across all Remainder-axis positions — the Ontological Shadow has the same cross-sectional area P at every position on the Remainder axis.

V.4 The 25-Pixel Tiling of the Unit Field

The reciprocal $1/P = 1/0.04 = 25$ was noted in Essay I as the number of non-overlapping Structural Pixels that tile the total unit field *[0,1]*. In the geometry of *NS*, this tiling has a precise construction.

The Source axis of *NS* has $n = 10$ lattice steps. The Boundary margin covers $\sigma / \delta = 4$ of these steps. Each Source-axis lattice step combined with the full Boundary-margin extent gives one Structural Pixel. Across all 10 Source-axis steps, 10 Structural Pixels can be placed. But the full unit field is two-dimensional (Source x Boundary projection), and 10 Source steps x 4 Boundary steps / $(\sigma / \delta) = 10 \times 4 / 4 = 10$ pixels in the Source direction is not the complete picture.

Derivation V.25 · **The 25-Pixel Tiling**

The Structural Pixel $P = \delta \times \sigma$ tiles the unit square $[0,1] \times [0,1]$ (Source x Boundary projection):

Number of pixels in Source direction:

$$1 / \delta = 1 / (1/10) = 10 \tag{7}$$

Number of pixels in Boundary direction:

$$1 / \sigma = 1 / (2/5) = 5/2 \Rightarrow \text{not integer} \tag{8}$$

The non-integer count in the Boundary direction reflects that $\sigma = 2/5$ does not divide 1 into a whole number of σ-width strips. Instead, the tiling is:

$$\text{Total unit field area} = 1 \times 1 = 1 \tag{9}$$

$$\text{Pixel area} = P = 1/25 = 0.04 \tag{10}$$

$$\text{Number of pixels} = 1 / P = 25 \tag{11}$$

Correct computation: the unit square $[0,1] \times [0,1]$ has area 1. Each Structural Pixel has area $P = \delta \times \sigma = (1/10) \times (2/5) = 2/50 = 1/25$. The number of non-overlapping pixels: $1 / (1/25) = 25$. This is exact and integer, confirming that 25 Structural Pixels tile the unit square perfectly.

Part VI — The Grand Geometric Partition

Parts I through V have established nothing's three-dimensional structural space NS, named its three axes, derived the Triadic Sub-Volume, derived the dimensional collapse and the Remainder margin, and located the Structural Pixel and the Ontological Shadow precisely within *NS*. Part VI brings these results together into the Grand Geometric Partition: the complete, exhaustive, exclusive partition of NS into its structurally distinct sub-regions.

VI.1 The Five Regions of Nothing's Structural Space

Nothing's structural space *NS = [0,1]^3* contains the following structurally distinct regions:

1. **The Triadic Sub-Volume (*TSV*):** the three-dimensional occupied region *[0,E] x [0,F] x [0,C] = [0,4/5] x [0,3/5] x [0,7/10]*. Volume = *TI = 42/125 = 0.336*. This is the region where all three primitives are simultaneously active — the structural content of nothing's inverted modes.

2. **The Source Margin Slab (*SMS*):** the region *[E,1] x [0,1] x [0,1] = [4/5,1] x [0,1] x [0,1]*. Volume = *I_min x 1 x 1 = 1/5 = 0.2*. This is the three-dimensional slab corresponding to the Source clearance margin — the region where the Source-axis coordinate is in the Source margin.

3. **The Boundary Margin Slab (*BMS*):** the region *[0,1] x [F,1] x [0,1] = [0,1] x [3/5,1] x [0,1]*. Volume = *1 x σ x 1 = 2/5 = 0.4*. This is the three-dimensional slab corresponding to the Boundary clearance margin — the region where the Boundary-axis coordinate is in the Boundary margin.

4. **The Remainder Margin Slab (*RMS*):** the region *[0,1] x [0,1] x [δ,1] = [0,1] x [0,1] x [0.7016,1]*. Volume = *1 x 1 x Λ = 0.2984*. This is the three-dimensional slab corresponding to the Remainder clearance margin.

5. The Interior Margins (*IMR*): the regions of *NS* that lie outside *TSV* and outside all three slabs. These are the geometric regions corresponding to axis extents between the primitive values and the margin beginnings — the non-*TSV*, non-margin portions of *NS*. Specifically: for each axis, the region between the axis's primitive extent and the opposite face's primitive extent.

However, the five-region description above contains overlaps. The three slabs are not mutually exclusive — their pairwise and triple intersections are non-empty. The partition requires inclusion-exclusion to produce a truly exclusive and exhaustive decomposition.

Derivation V.GGP · **The Grand Geometric Partition by Inclusion-Exclusion**

Let the three clearance slabs be:

$$SMS = [E,1] \times [0,1] \times [0,1] \quad Vol = I\_min = 1/5 = 0.200 \tag{1}$$

$$BMS = [0,1] \times [F,1] \times [0,1] \quad Vol = \sigma = 2/5 = 0.400 \tag{2}$$

$$RMS = [0,1] \times [0,1] \times [\delta,1] \quad Vol = \Lambda = 0.2984 \tag{3}$$

Pairwise intersections:

$$SMS \cap BMS = [E,1] \times [F,1] \times [0,1]$$
$$Vol = I\_min \times \sigma = (1/5)(2/5) = 2/25 = 0.080 \tag{4}$$

$$SMS \cap RMS = [E,1] \times [0,1] \times [\delta,1]$$
$$Vol = I\_min \times \Lambda = (1/5)(0.2984) = 0.0597 \tag{5}$$

$$BMS \cap RMS = [0,1] \times [F,1] \times [\delta,1]$$
$$Vol = \sigma \times \Lambda = (2/5)(0.2984) = 0.1194 \tag{6}$$

Triple intersection:

$$SMS \cap BMS \cap RMS = [E,1] \times [F,1] \times [\delta,1]$$
$$Vol = I\_min \times \sigma \times \Lambda = 0.0240 \tag{7}$$

By inclusion-exclusion, the union of the three slabs:

$$Vol(SMS \cup BMS \cup RMS) \tag{8}$$

$$= Vol(SMS) + Vol(BMS) + Vol(RMS) \tag{9}$$

$$- Vol(SMS \cap BMS) - Vol(SMS \cap RMS) - Vol(BMS \cap RMS) \tag{10}$$

$$+ Vol(SMS \cap BMS \cap RMS) \tag{11}$$

$$= 0.200 + 0.400 + 0.2984 - 0.080 - 0.0597 - 0.1194 + 0.0240 \tag{12}$$

$$= 0.8984 - 0.2591 + 0.0240 \tag{13}$$

$$= 0.6633 \tag{14}$$

The Triadic Sub-Volume TSV has volume TI = 42/125 = 0.336. Verification:

$$Vol(TSV) + Vol(SMS \cup BMS \cup RMS) \text{ should account for all of } NS \tag{15}$$

$$0.336 + 0.6633 = 0.9993 \text{ (close to 1 but not exact due to the non-lattice } \Lambda) \tag{16}$$

The small deficit 1 - 0.9993 = 0.0007 is the geometric expression of the sub-lattice residual $\varepsilon\_R = 0.0016$ accumulated through the intersection volumes. The perfect partition is:

$$Vol(TSV) + Vol(\text{complement of } TSV \text{ in } NS) = TI + (1 - TI) = 1 = \Omega \text{ ok} \tag{17}$$

VI.2 T.GGP — The Grand Geometric Partition

The complete description of *NS* as a partitioned structural space is now available. The fundamental partition is binary: the Triadic Sub-Volume (occupied territory) and its complement (clearance territory). The clearance territory is then sub-divided by the three clearance margins and their intersections.

Theorem T.GGP · **The Grand Geometric Partition**

Nothing's structural space $NS = [0,1]^3$ is partitioned into:

Primary partition:

```
TSV: the Triadic Sub-Volume  — occupied territory
  — Vol = TI = 42/125 = 0.336                                   (18)

CC:  the Clearance Complex  — unoccupied territory
  — Vol = 1 - TI = 83/125 = 0.664                               (19)
```

The Clearance Complex CC is further structured by three clearance margins:

```
SM: Source margin  (axis [E,1] on x_S)  — I_min = ⅕             (20)

BM: Boundary margin  (axis [F,1] on x_B)  — σ = ⅖               (21)

RM: Remainder margin  (axis [δ,1] on x_R)  — Λ = 0.2984          (22)
```

The Ontological Shadow $OS = [E,E+\delta] \times [F,1] \times [0,1]$ is the geometric intersection of the first Source-margin step with the full Boundary-margin slab. $Vol(OS) = P = 1/25 = 0.04$.

Every position in *NS* belongs to either *TSV* or *CC*. Every position in *CC* belongs to at least one of the three margin slabs. The partition is exhaustive (every position accounted for) and complete (no position requires information beyond the six constants: E, F, C, δ, β, Ω — all derived in Essay I with zero free parameters).

VI.3 The Geometric Hierarchy of the Three Clearances

The three clearance margins have sizes that reflect the hierarchy established in Essay I (P.HI): Boundary clearance > Remainder clearance > Source clearance. In the geometry of *NS*:

Principle P.GH · **The Geometric Hierarchy of Clearances**

Clearance margin volumes on their respective axes:

```
BM (Boundary):     σ  = 2/5   = 0.4000
(4 lattice steps — maximum)
```
(23)

```
RM (Remainder):   Λ = 0.2984
(2.984 steps — intermediate)
```
(24)

```
SM (Source):       I_min  = 1/5   = 0.2000
(2 lattice steps — minimum)
```
(25)

Strict ordering: $\sigma > \Lambda > I_{min} > 0$ $(0.4 > 0.2984 > 0.2 > 0)$. This ordering is the geometric reading of P.HI (Essay I): the Boundary margin is largest because the Boundary face is the differentiating act requiring maximum clearance; the Source margin is smallest because the Source face is the reservoir requiring minimum interior extent; the Remainder margin is intermediate because the Remainder face is the mediating structure.

The geometric hierarchy is not imposed on the space — it is derived from the forcing conditions of T.UC and the dimensional collapse. It is the spatial expression of nothing's structural asymmetry.

VI.4 The Geometric Failure Modes

Essay I derived three failure modes algebraically: the Deterministic Crystal ($\sigma = 0$), Total Dissolution (σ approaching 1), and Causal Closure Seizure ($\Lambda = 0$). Part VI gives each its geometric reading within *NS*.

VI.4a The Deterministic Crystal as Geometric Collapse

The Deterministic Crystal occurs when $\sigma = 0$ and $F = 1$. In the geometry of *NS*, this means: the Boundary margin $BM = [F, 1] = [1, 1]$ shrinks to a single point — the boundary position $x\_B = 1$. The Boundary-margin slab *BMS* collapses to a two-dimensional face of $NS: \{x\_B = 1\} \times [0,1] \times [0,1]$. The slab has zero volume. The Structural Pixel $P = \delta \times \sigma = \delta \times 0 = 0$. The Ontological Shadow has zero volume. Nothing's geometric clearance on the Boundary axis has been completely consumed. The Triadic Sub-Volume in this limit expands to $[0,1] \times [0,1] \times [0,C]$ — a flat slab with no Boundary-axis clearance. *NS* is no longer a three-dimensional structure that maintains the Boundary face as a separate geometric region.

VI.4b Total Dissolution as Geometric Expansion

Total Dissolution occurs when *F* approaching *0* and *σ* approaching *1*. In *NS*, the Boundary margin *BM* approaches $[0, 1]$ — the full Boundary axis. The Triadic Sub-Volume in the limit has zero Boundary-axis extent *([0, F] = [0, 0] = a single point)*. The *TSV* collapses to a two-dimensional face of $NS: [0,E] \times \{x\_B = 0\} \times [0,C]$. The three-dimensional structure loses the Boundary dimension entirely. With $F = 0$, the Registration density occupies no extent on the Boundary axis — the Boundary-axis contribution to *TI* is zero, making $TI = 0$. The dimensional collapse then gives $\delta = 0^\beta = 0$, and $\Lambda = 1$. The entire Remainder axis is clearance. Nothing's structural space has no occupied sub-volume — it has dissolved into clearance on all axes.

VI.4c Causal Closure Seizure as Remainder-Axis Saturation

Causal Closure Seizure occurs when $\delta = 1$ and $\Lambda = 0$. In *NS*, the Remainder margin $RM = [1, 1]$ shrinks to the single point $x\_R = 1$. The entire Remainder axis is occupied by the dimensional collapse result. There is no geometric residual on the Remainder axis. The Remainder face's clearance has been consumed. In the geometry of *NS*, this means the power-law projection of the Triadic Sub-Volume onto the Remainder axis fills the Remainder axis completely — leaving no unoccupied territory. The structural space has no Remainder margin, and the Triad cannot maintain the Remainder face as a separate geometric region distinct from its complement.

Conclusion

What Has Been Derived

Essay II has derived nothing's three-dimensional structural space from the logical results of Essay I. No external framework was imported. No new axiom was introduced. Every result follows from the logical architecture already established.

Part I: D.LP identified the logical position with the lattice step $\delta = 1/10$. T.GA (the Axis Theorem) proved that mutual exclusion and exhaustion at every logical position requires each face to vary along its own independent axis — sharing an axis forces either gap positions (violating exhaustion) or adjacent-step boundary violations (violating T.DI1). T.3D followed directly: three faces, three axes, $d = 3$. Reductio confirmed: $d = 1$ and $d = 2$ are foreclosed by pigeonhole and T.GA; $d > 3$ is foreclosed because empty axes violate exhaustion.

Part II: D.3A named the three axes (Source, Boundary, Remainder) and bounded each by $[0,1]$ (from T.FU). D.LS established the $10x10x10$ discrete lattice of 1331 points, with the structurally active interior having 729 positions (strictly between 0 and 1 on all axes). D.NS defined nothing's structural space as the unit cube $[0,1]$^3 discretised at $\delta = 1/10$. P.CPG gave T.CP its geometric reading: the co-primary partition of each axis into primitive and clearance intervals.

Part III: The Source margin $SM = [4/5, 1]$ — length $1/5$, two lattice steps, one interior point — was derived as the minimum interior extent for Source-Boundary distinction. The Boundary margin $BM = [3/5, 1]$ — length $2/5$, four lattice steps, three interior points — was derived as the dual-foreclosed unique interior value of the Registration clearance. The Remainder margin $RM = [0.7016, 1]$ — length $\Lambda = 0.2984$, 2.984 non-integer lattice steps — was identified as a three-dimensional residual requiring the dimensional collapse of Part IV.

Part IV: D.TSV defined the Triadic Sub-Volume as the rectangular sub-cube *[0,E] x [0,F] x [0,C]* with volume $TI = 42/125$. T.DC derived the dimensional collapse: the power law $\delta = TI^\beta$ maps the three-dimensional *TSV* to the Remainder-axis position $\delta = 0.7016$. V.DCG computed the collapse residual $\delta - C = 0.0016$ as the non-linear excess of the three-dimensional projection over the naive one-dimensional Constraint complement. T.RM established the Remainder margin as a function of all three axes simultaneously via the dimensional collapse.

Part V: D.BMS defined the Boundary-margin slab as the three-dimensional region *[0,1] x [F,1] x [0,1]* of volume $\sigma = 0.4$. D.SP.II derived the Structural Pixel as the cross-sectional area of this slab at one Source-axis lattice step: $P = \delta \times \sigma = 1/25 = 0.04$. D.OS located the Ontological Shadow precisely within NS: the three-dimensional region *[E, E+δ] x [F,1] x [0,1]* where the Source clearance is at its minimum entry and the Boundary clearance is at full extent. V.25 confirmed *25* non-overlapping Structural Pixels tile the unit square.

Part VI: V.GGP derived the Grand Geometric Partition by inclusion-exclusion. T.GGP stated the complete partition: *TSV* (occupied, $vol = 0.336$) plus the Clearance Complex *CC* (unoccupied, $vol = 0.664$), with CC structured by three clearance margin slabs and their intersections. P.GH gave the hierarchy $\sigma > \Lambda > I\_min$ its geometric reading. VI.4 derived the three failure modes as geometric conditions: the Deterministic Crystal as Boundary-axis slab collapse, Total Dissolution as Boundary-dimension disappearance, Causal Closure Seizure as Remainder-axis saturation.

The Terminal Result

Nothing's structural space is not a container. It is the geometric expression of nothing's logical architecture: three independent axes, forced by T.DIM and T.GA; bounded by $\Omega = 1$, forced by T.FU; discretised at $\delta = 1/10$, forced by V.DS; occupied by the Triadic Sub-Volume $TI = 42/125$, forced by T.UC and C.IV; and clearanced by three margins whose positions and sizes are forced by the same six constraints that determined the primitives.

Every position in *NS* is structurally accounted for. Every constant is derived. Nothing's three-dimensional structural space *[0,1]^3* is the complete geometric architecture of the Triad — sealed, zero free parameters, and entirely internal to the logical structure that Essay I established from absolute nothing.

Theorem T.TA · **The Terminal Architecture**

Nothing's structural space is the *10x10x10* discrete unit cube *NS = [0,1]^3*, derived from:

$$d = 3\text{: three independent axes, forced by } T.DIM + T.GA + T.3D \tag{26}$$

$$\Omega = 1\text{: axis ceiling, forced by T.FU (Essay I)} \tag{27}$$

$$\delta = 1/10\text{: lattice grain, forced by Diophantine search V.DS (Essay I)} \tag{28}$$

$$TSV = [0,E]x[0,F]x[0,C]\text{: Triadic Sub-Volume, forced by T.UC (Essay I)} \tag{29}$$

$$Vol(TSV) = TI = 42/125\text{: three-dimensional volume, forced by C.IV (Essay I)} \tag{30}$$

$$SM = [4/5,1] \text{ on } x\_S\text{: Source margin, length } I\_min = 1/5 \text{ (V.MB, Essay I)} \tag{31}$$

$$BM = [3/5,1] \text{ on } x\_B\text{: Boundary margin, length } \sigma = 2/5 \text{ (P.RB, Essay I)} \tag{32}$$

$$RM = [0.7016,1] \text{ on } x\_R\text{: Remainder margin, length } \Lambda = 0.2984 \text{ (V.OP, Essay I)} \tag{33}$$

$$OS = [E,E+\delta]x[F,1]x[0,1]\text{: Ontological Shadow, area } P = 1/25 \text{ (V.SP, Essay I)} \tag{34}$$

Nothing's three-dimensional structural space is the geometric necessity of nothing's logical self-consistency. It requires no external spatial container. It is what nothing is, seen geometrically.

Bibliography

The following works are cited for structural isomorphism confirmation only. None serve as derivational foundations for the present framework. All results herein are derived from the internal logic of the Triad and the normalised field alone.

Internal series references

Pretorius, E. (2026). *Gradientology - Foundations of the Primordial Triad*: Treatises I-XVI. Zenodo.

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ADDENDUM: UNIVERSAL TERMS OF ENGAGEMENT FOR THE RESIDUAL SUITE

Preamble This addendum serves as the binding and immutable interpretive constraint specifically for the **Residual Suite** (Validation Protocols and Ledger Audits for the 20% Minimum Bit-Budget, 30% Ontological Gap, and 40% Structural Volatility). Its purpose is to formalize the absolute universality of these geometric clearances, explicitly foreclosing any attempt to restrict them to a specific domain, timeframe, or material.

CONSTRAINT 1: Phase Invariance (The Permanent Mechanics)

The three residuals ($I\_min \approx 0.2, \Lambda \approx 0.298, \sigma = 0.4$) are not pre-kinetic artifacts or temporary conditions of the primordial stasis. They are the permanent, load-bearing mechanical clearances of the universe. Analysts are strictly prohibited from treating these gaps as confined to Phase I. They operate continuously across all phases—preventing the Deterministic Crystal in Phase II and acting as the Velocity/Saturation blocks that mandate perpetual novelty in Phase III. The gaps are the permanent operating tolerance of the engine.

CONSTRAINT 2: Scale Invariance (The Scalar Mandate)

The structural clearances apply with absolute isomorphic fidelity at every scalar resolution of the **Veldt.** Readers and analysts are prohibited from treating the residuals as exclusively "quantum" phenomena or exclusively "cosmological" phenomena. The 40% Structural Volatility is identically the source of Heisenberg's Uncertainty Principle at the micro-scale, the mandated tolerance in macroscopic precision engineering, and the limit of resolution in computational mathematics. The math is scale-invariant.

CONSTRAINT 3: Substrate Neutrality (The Purge of Material Bias)

The residuals are pure mathematical inevitabilities of a discrete lattice. They must never be reified by attributing them to a specific material substrate. The 20%, 30%, and 40% clearances govern biological ecosystems, artificial intelligence networks, silicon processing, and the astrophysical void equally. Importing material, biological, or anthropic properties to explain the existence of these gaps is a prohibited circularity.

CONSTRAINT 4: Epistemic Foreclosure (Anti-Reification of Uncertainty)

Regardless of the phase, scale, or substrate being analyzed, no reader may interpret these three residuals as measurement errors, statistical noise, or "epistemic limits" arising from observer ignorance. They are derived **Mechanical Necessities**. To claim these gaps can be "closed" by better measurement, or to treat them as probabilistic noise, is to fundamentally misunderstand the engine. **The residuals are not what the engine lacks — they are what the engine is.**

CONSTRAINT 5: Final Demarcation and Logical Necrosis

These Terms of Engagement are non-negotiable. Any attempt to localize these clearances to a specific phase, restrict them to a specific scale, tie them to a specific substrate, or dismiss them as probabilistic contingencies constitutes a formal epistemic breach classified as **Logical Necrosis**.